# Clinical Coding, Classification and Documentation Improvement

*How the record becomes a code, why the code decides the payment, and why only the clinician can close the gap.*

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| **Module** | 09 of 12 |
| **Study time** | 100 minutes |
| **Audience** | Coders, medical records staff, clinicians, billing and insurance teams, CDI staff |
| **Assessment** | Knowledge check plus a 50-question scored bank |
| **Standards basis** | ICD-10 and ICD-10-PCS/ICD-9-CM conventions; NABH 6th Edition IMS chapter; PMJAY and insurance package requirements |

Coding converts a clinical narrative into a classification that can be counted, compared and paid against. The conversion is rule-bound and the rules are strict: a coder may assign only what the record supports. That single constraint explains why under-documented hospitals are under-paid, why coders raise queries, and why clinical documentation improvement is a clinical activity rather than a billing one.

**LEARNING OUTCOMES — by the end of this module you will be able to:**

- Describe the path from clinical care to a coded claim and name what happens at each step
- Read an ICD-10 code and explain what each part of it conveys
- Distinguish principal diagnosis, secondary diagnosis, comorbidity and complication
- Identify the documentation phrases that make a record uncodeable and state what should be written instead
- Write a compliant clinician query and recognise a leading one
- Explain the difference between under-coding, over-coding and up-coding, and the consequences of each

## 9.1 From bedside to bill




*Figure 9.1 — Where coding sits between clinical care and the claim.*

The crucial property of this chain is that coding adds nothing. The coder reads the record and assigns codes that the record supports. Where the record is ambiguous, incomplete or internally contradictory, the coder may not resolve it by clinical inference; they must ask the clinician. This is not bureaucratic caution. A coder who infers a diagnosis is manufacturing a clinical statement that no doctor made, which is both clinically wrong and, where it increases payment, fraudulent.

## 9.2 Reading a code

**Reading an ICD-10 code**




| Term in the record | Why it matters | What the coder needs |
|---|---|---|
| Principal diagnosis | Drives the package and the length-of-stay benchmark | The condition, established after study, chiefly responsible for the admission |
| Secondary diagnosis | Reflects complexity and justifies resource use | Conditions that coexisted and affected management this admission |
| Complication | Distinguishes expected course from adverse event | Stated as arising during or after care, with the link made explicit |
| Comorbidity | Raises the expected cost and risk-adjusts outcomes | Pre-existing condition that increased the care needed |
| Suspected / rule-out | Coded differently in inpatient and outpatient settings | Whether it was confirmed, excluded or still suspected at discharge |

*Figure 9.2 — The structure of an ICD-10 code, and the diagnosis types a coder must distinguish.*

ICD-10 codes have a chapter letter, a two-digit category, and further characters giving specificity. The further characters are where documentation quality shows: a record that says ‘diabetes’ supports only an unspecified code, while a record that says ‘type 2 diabetes mellitus with diabetic nephropathy, on insulin’ supports a specific one. The specific code reflects the real complexity of the patient, justifies the resources used, and survives payer scrutiny.

Procedure classification varies by hospital: ICD-9-CM volume 3, ICD-10-PCS, or a national or scheme-specific procedure list such as those used under PMJAY. The documentation requirement is the same in all of them — the procedure must be named in full, with approach, site, laterality and any device used.

## 9.3 The four diagnosis types

| Type | Definition | Why it matters | What the record must state |
|---|---|---|---|
| Principal diagnosis | The condition established after study to be chiefly responsible for the admission | Drives the package, the expected length of stay and the benchmark comparison | A definite final diagnosis, not the presenting symptom and not the admitting provisional |
| Secondary diagnosis | Conditions that coexisted and affected patient management during this admission | Reflects complexity and justifies resource use and longer stay | That the condition was actively managed, monitored, investigated or extended the stay |

| Type | Definition | Why it matters | What the record must state |
|---|---|---|---|
| Comorbidity | A pre-existing condition that increased the care required | Risk-adjusts outcome comparisons and supports higher packages | The condition, its severity, and how it affected management this admission |
| Complication | A condition arising during or as a result of care | Distinguishes an expected course from an adverse event, and feeds safety indicators | An explicit causal link: that the condition arose during or because of the care given |

> **CAUTION**
> The distinction between a comorbidity and a complication is clinical and must be made by the clinician. A coder who guesses will either understate the patient's complexity or record an adverse event that never occurred, and both appear in the hospital's outcome data.

## 9.4 The phrases that make a record uncodeable

| Written in the record | Why it cannot be coded well | Write instead |
|---|---|---|
| 'Fever for evaluation' as the final diagnosis | A symptom, not a diagnosis; no principal diagnosis can be assigned | The condition established, or if genuinely undetermined, state that and record what was excluded |
| 'Anaemia' | No type, no severity, no cause | 'Severe iron-deficiency anaemia, haemoglobin 6.1 g/dL, transfused two units' |
| 'Pneumonia' | No organism, no site, no severity | 'Community-acquired right lower lobe pneumonia, CURB-65 score 3' |
| 'Renal failure' | Acute or chronic, stage, cause all missing | 'Acute kidney injury stage 2 on chronic kidney disease stage 3b, secondary to sepsis' |
| 'Sepsis?' or '? sepsis' | A question mark leaves the status undetermined | State whether sepsis was confirmed, excluded, or treated as suspected and on what basis |
| 'CVA' | Ischaemic or haemorrhagic, territory, and whether acute, all missing | 'Acute ischaemic stroke, left middle cerebral artery territory, thrombolysed' |
| 'Diabetes' | Type, control, and complications missing | 'Type 2 diabetes mellitus with hyperglycaemia and diabetic peripheral neuropathy' |
| 'Post-op fever' | Neither a diagnosis nor a stated complication | The cause if known, or an explicit statement that it is being investigated, with what has been excluded |
| 'Shifted to ICU' with no reason | No clinical justification for the higher level of care | The clinical indication for intensive care, with the parameters that triggered it |

## 9.5 The clinician query

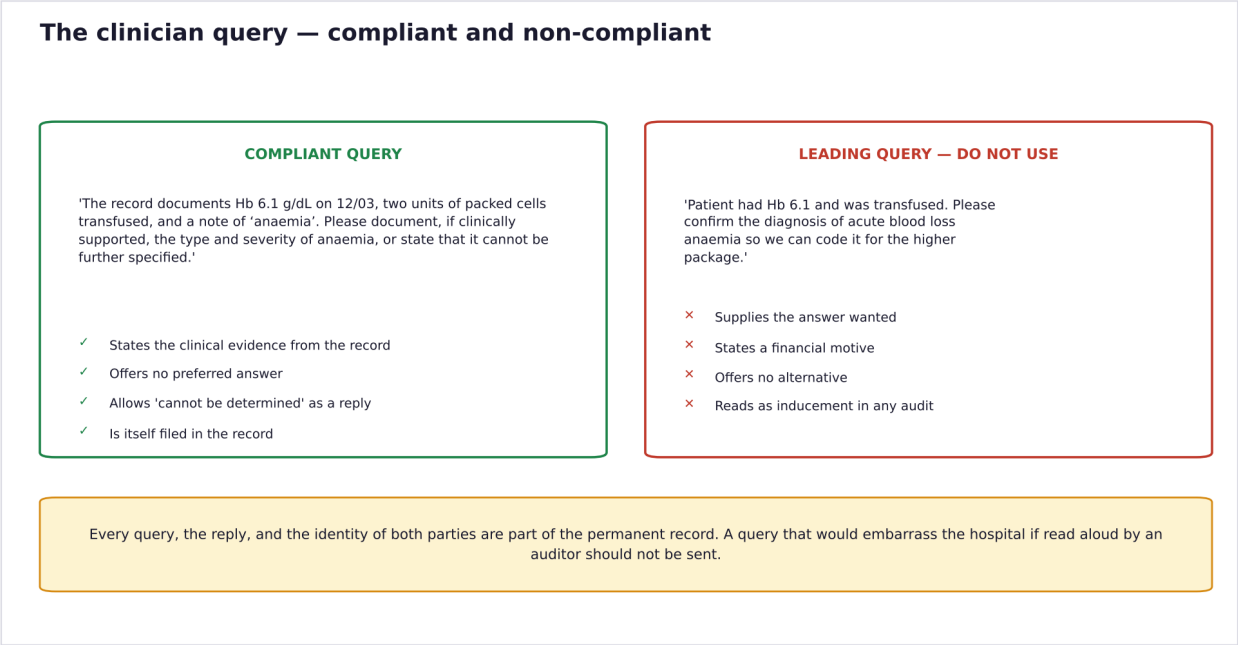



*Figure 9.3 — A compliant query and a leading one.*

The query exists because the coder cannot infer. A compliant query states the clinical evidence found in the record, asks the clinician to document further if clinically supported, offers no preferred answer, explicitly allows 'cannot be determined' as a reply, and is itself filed in the record along with the answer.

- Quote the clinical indicators from the record — values, treatments given, monitoring undertaken
- Ask an open question about the clinical significance, not a yes/no question about a specific diagnosis
- Never mention reimbursement, package rates, the effect on the claim, or the hospital's performance data
- Offer multiple reasonable options where options are offered, including 'other' and 'cannot be clinically determined'
- Set a response timeframe and track unanswered queries
- File the query and the response in the medical record; both are part of it

> **CAUTION**
> A query that would embarrass the hospital if read aloud by an auditor should not be sent. Queries and responses are discoverable, and a pattern of leading queries is evidence of systematic up-coding regardless of what any individual record shows.

## 9.6 Under-coding, over-coding and up-coding

| | What it is | Cause | Consequence |
|---|---|---|---|
| Under-coding | Assigning fewer or less specific codes than the record supports | Coder under time pressure, poor documentation, incomplete abstraction | Lost revenue, understated case complexity, poor-looking outcome data because severity is not captured |
| Over-coding | Assigning codes the record does not support | Coder inference, template documentation that asserts findings never made | Claim rejection, repayment demands, and in a pattern, an allegation of fraud |
| Up-coding | Deliberately selecting a higher-paying code or sequence | Targets, incentives, leading queries, pressure on coders | Fraud finding, de-empanelment from schemes, criminal exposure |

| | What it is | Cause | Consequence |
|---|---|---|---|
| Unbundling | Coding components separately where a combination code exists | Misunderstanding of coding conventions, or deliberate | Claim rejection and audit scrutiny of the whole account |

> **KEY POINT**
> Under-coding and over-coding are both documentation failures before they are coding failures. The fix for both is the same: a record that states exactly what the clinician found, concluded, and did — no more and no less.

## 9.7 Clinical documentation improvement

A clinical documentation improvement programme reviews records concurrently — while the patient is still in hospital — rather than after discharge, because a clarification sought on day two is answered from the patient in front of the clinician, while the same question asked six weeks later is answered from memory or not at all.

1. Identify the records to review: high length of stay, intensive care, high-value packages, cases with complications, and specialties with known documentation gaps.
2. Review against clinical indicators: a low haemoglobin with no anaemia diagnosis, a positive culture with no infection diagnosis, an elevated creatinine with no renal diagnosis, vasopressor use with no shock diagnosis.
3. Query concurrently, compliantly, and to the clinician who is actually looking after the patient.
4. Track the response rate and the agreement rate, which measure whether the programme is working.
5. Feed the patterns back as education by specialty, not as individual criticism.
6. Measure the effect on case-mix index, on claim rejection rates and on the completeness of coded comorbidities.

## 9.8 Coding and the insurance claim

Payers reject claims for documentation reasons far more often than for clinical ones. The recurring grounds are predictable and almost all are fixable at the point of care.

| Rejection ground | What the record lacked | Fix at the point of care |
|---|---|---|
| Diagnosis not supported | Final diagnosis absent, or a symptom recorded as the diagnosis | State the final diagnosis on the diagnosis sheet and the summary, with specificity |
| Length of stay not justified | No daily progress notes, or notes that do not show why the patient still needed inpatient care | A daily note stating the clinical reason the patient remains an inpatient |
| Procedure not documented | Operative note missing, illegible, or lacking the procedure name | Operative note immediately after the procedure, with the full procedure name |
| Implant not evidenced | No lot number, no sticker, no invoice linkage | Implant stickers in the record and details in the operative note |
| Higher level of care not justified | No documented indication for intensive care | The clinical indication and the triggering parameters recorded at the time of transfer |
| Investigation not justified | Test ordered with no clinical question recorded | A line in the notes stating why the test is being done |
| Consent missing or incomplete | Blank fields or the wrong form | Complete the correct procedure-specific form and file it with the episode |
| Pre-existing condition not disclosed | Comorbidity undocumented at admission | A complete past history at admission, including duration of chronic conditions |

## 9.9 Coding ethics

Coders occupy an uncomfortable position: they are measured on productivity and accuracy, and they are sometimes pressed to produce a particular result. Three principles protect both the coder and the hospital. Code only what the record supports. Query rather than infer. Escalate pressure to change a code rather than accommodate it, and keep a record of having done so.

## 9.10 Hospital statistics derived from the record

The medical records department is the source of the hospital's activity statistics, and those statistics are only as good as the abstraction behind them. Senior management decisions about staffing, beds and services rest on these numbers.

| Statistic | How it is calculated | What corrupts it |
|---|---|---|
| Bed occupancy rate | Inpatient days of care / (available beds × days in the period) × 100 | Inconsistent counting of day cases, and beds counted as available when they are closed |
| Average length of stay | Inpatient days of care / discharges including deaths | Day cases included or excluded inconsistently; long-stay outliers not reported separately |
| Bed turnover | Discharges including deaths / available beds | Transfers between wards counted as discharges |
| Turnover interval | (Available bed days – occupied bed days) / discharges including deaths | Occupancy counted at a single census hour rather than across the day |
| Gross mortality rate | Deaths / discharges including deaths × 100 | Deaths within a defined period of admission excluded or included inconsistently |
| Caesarean section rate | Caesarean deliveries / total deliveries × 100 | Denominator defined differently between departments |
| Day-case rate | Day cases / (day cases + inpatient episodes for the same procedures) | Inconsistent classification of overnight stays |

Definitions must be written down and applied consistently across periods, because a change in definition produces an apparent change in performance that is entirely artefactual. Where a definition changes, the records department should say so when the statistic is reported.

## 9.11 Statutory and registry reporting

Several reporting obligations are fed from the medical record and the records department is usually the point of failure when they are missed.

| Report | Trigger | Documentation needed |
|---|---|---|
| Notifiable disease reporting | Diagnosis of a disease on the state's notifiable list | The diagnosis, the date of notification, to whom, and the acknowledgement |
| Births and deaths registration | Every birth and death in the institution | Statutory forms completed and submitted within the prescribed period, with counterfoils retained |
| Cancer registry | A new diagnosis of malignancy | Site, morphology, stage, basis of diagnosis, date of incidence, and treatment given |
| Pharmacovigilance | A suspected adverse drug reaction | The reaction, the suspected drug, causality assessment, outcome, and the report reference |
| Medical device adverse events | Device malfunction or injury | Device details including lot number, the event, the outcome, and the report to the manufacturer and the regulator |
| Haemovigilance | A transfusion reaction | The unit details, the reaction, the investigation, and the report |

| Report | Trigger | Documentation needed |
|---|---|---|
| Maternal and infant death surveillance | A maternal or infant death | Case review documentation and the prescribed reporting format |

## 9.12 Coding quality and productivity

Coding accuracy is audited the way any other process is: a sample re-coded independently and compared, with disagreements classified. Productivity targets without an accuracy audit produce fast, wrong coding.

- Audit a defined sample monthly, re-coded blind by a second coder or an external reviewer
- Classify disagreements: principal diagnosis changed, secondary diagnoses missed, procedure code wrong, sequencing wrong
- Separate coder error from documentation inadequacy, because the remedies are different
- Track the proportion of cases where the documentation prevented accurate coding; that is the number clinicians need to see
- Review coder queries for compliance as well as for response rate
- Treat a change in case-mix index without a change in case mix as a signal requiring investigation

## 9.13 Package-based schemes and the documentation they require

Publicly funded schemes and insurance packages pay against a defined package, and each package specifies the documentation that must accompany the claim. The commonest reasons for rejection are documentary rather than clinical.

- Pre-authorisation where the scheme requires it, with the clinical justification recorded
- Identity and eligibility verification documented at admission
- The procedure recorded in the exact terms the package uses, so that the mapping is unambiguous
- Implant invoices and stickers where the package includes an implant component
- Photographic or other evidence where the scheme specifies it, obtained with consent
- Discharge summary containing the elements the scheme lists, which are often more prescriptive than the hospital’s own format
- Length of stay justified by daily documentation where the package has a defined stay

> **KEY POINT**
> Build the scheme’s documentary requirements into the clinical workflow rather than assembling them after discharge. A photograph that must be taken during the procedure cannot be obtained six weeks later, and the claim fails for a reason that had nothing to do with the care.

## 9.14 Terminology, synonyms and why precision matters

Clinicians use synonyms freely and classifications do not. The same condition written three ways produces three coding outcomes, and a condition written loosely produces none.

| Written loosely | The problem | Written precisely |
|---|---|---|
| Heart attack | Not a classification term; type, site and timing missing | Acute ST-elevation myocardial infarction, anterior wall |
| Stroke | Ischaemic or haemorrhagic changes everything downstream | Acute ischaemic stroke, left middle cerebral artery territory |
| Kidney problem | Acute, chronic or structural all code differently | Chronic kidney disease stage 4, secondary to diabetic nephropathy |

| Written loosely | The problem | Written precisely |
|---|---|---|
| Infection | Site and organism missing | Catheter-associated urinary tract infection, Escherichia coli, confirmed on culture |
| Breathlessness | A symptom, not a diagnosis | Acute exacerbation of chronic obstructive pulmonary disease, type 2 respiratory failure |
| Low platelets | A finding, not a diagnosis | Immune thrombocytopenia, platelet count 18,000 |
| Septicaemia used loosely | A legacy term; sepsis has clinical criteria | Sepsis secondary to community-acquired pneumonia, meeting the hospital's sepsis criteria |

Hospitals moving towards structured electronic records will increasingly select diagnoses from a terminology such as SNOMED CT rather than typing them. That removes the synonym problem but introduces a selection problem: picking the nearest-looking term from a list is as damaging as writing a vague one, and the selected term must be the one the clinician means.

## 9.15 Abstracting: what is captured besides codes

Coding is one part of abstraction. The department also captures the data on which the hospital's reporting depends, and an error here propagates into every subsequent report.

- Admission and discharge dates and times, and the discharge disposition — home, transferred, against advice, absconded, died
- The treating unit and consultant, which determines whose outcomes these are
- Procedures with dates, which determines the procedure counts and the theatre utilisation data
- Transfers between units within the admission, which affect intensive care occupancy and length-of-stay calculations
- The payer and scheme, which drives the revenue analysis
- Readmission linkage to the earlier episode, which is impossible without a reliable unit number
- Deaths within defined periods of admission or of a procedure, which feed the mortality indicators

## 9.16 Where documentation fails, chapter by chapter

| Area | The documentation gap that recurs |
|---|---|
| Infectious disease | Organism and site not stated; 'sepsis' used without the criteria being documented; source of infection not identified |
| Neoplasms | Primary site, morphology and whether the admission was for the primary, a secondary or a complication of treatment all left unclear |
| Endocrine and metabolic | Diabetes recorded without type, control status or complications; electrolyte abnormalities recorded as values without a diagnosis |
| Circulatory | Heart failure without type or severity; hypertension recorded as a background note rather than a managed condition |
| Respiratory | Pneumonia without site or acquisition setting; respiratory failure not documented despite oxygen and blood gas evidence |
| Injury and poisoning | External cause not recorded; the place and activity at the time of injury missing; intent not stated |
| Complications of care | The causal link not made explicit, so an adverse event is coded as a coincidental condition or vice versa |
| Obstetrics | Episode type, gestational age and the indication for intervention not stated together |

## 9.17 Working with your coder

The relationship between clinicians and coders determines how much of a hospital's work is visible in its data. A few practical habits remove most of the friction.

1. Write the final diagnosis on the diagnosis sheet and in the discharge summary, and make them agree.
2. List comorbidities and complications separately rather than in a single sentence, and say which is which.
3. Where a condition was suspected and excluded, say so explicitly; it is as useful as a positive finding.
4. Answer queries within the stated period, from the record and from your clinical knowledge, and document the answer in the record rather than only in an email.
5. Treat a query as a prompt about the record rather than a challenge to your clinical judgment; it almost always is.
6. Ask your coding team once a quarter which three documentation gaps they query most often in your specialty, and fix those three.

> **KEY POINT**
> A clinician who answers coder queries promptly and documents specifically is not doing the billing department a favour. They are making the hospital's data describe the patients they actually treated, which is what every outcome comparison, every benchmark and every resource argument then rests on.

## 9.18 Putting this module to work

1. Take five discharge summaries and check whether each states a final diagnosis with specificity, or a symptom.
2. Find a patient on your ward with a haemoglobin below 8 and check whether a diagnosis of anaemia with type and severity appears anywhere in the record.
3. Ask your coding team for the three documentation gaps they query most often in your specialty.
4. Look at your hospital's last month of claim rejections and count how many were documentation grounds rather than clinical ones.
5. Read one query your coding team sent last week. Was it compliant, or did it name the answer it wanted?

# Knowledge Check

Work through each item before reading the model answer. These are applied questions: the answer is the standard of documentation your assessor, your medico-legal adviser or the next clinician to read the file would expect.

Q1. **A record shows haemoglobin 6.1 g/dL, two units of packed cells transfused, and a progress note saying 'anaemia'. May the coder assign severe anaemia?**

**Answer:** No. The coder may assign only what the record states, and the record states 'anaemia' without type or severity. The coder must raise a compliant query citing the haemoglobin value and the transfusion, and asking the clinician to document the type and severity if clinically supported, or to state that it cannot be further specified. If the clinician documents it, the specific code follows; if not, the unspecified code stands.

Q2. **A hospital's case-mix index rises sharply after a new coding incentive is introduced, with no change in the patient population. What should be investigated?**

**Answer:** Whether the rise reflects better documentation or up-coding. The investigation should examine query patterns for leading questions, compare coded comorbidities against clinical indicators in the record, audit a random sample against source documentation, and look at whether specific high-value codes rose disproportionately. An incentive tied to coded severity is itself a control weakness and should be reconsidered.

Q3. **A clinician writes 'post-op fever' on day two after a bowel resection and nothing further. What are the coding and clinical consequences?**

**Answer:** Coding cannot determine whether this is an expected post-operative pyrexia, a complication such as a surgical site infection or an anastomotic leak, or an unrelated condition. The hospital therefore either loses the complexity entirely or risks recording a complication that was not confirmed. Clinically, the next reader has no idea what was considered or excluded. The note should state the differential, the investigations ordered and why, what has been excluded, and the working conclusion.

Q4. **A payer rejects a claim on the ground that the seven-day stay is not justified, even though the patient was genuinely sick throughout. What was probably missing?**

**Answer:** Daily progress notes that state why the patient still required inpatient care. A file with a good admission note and a good discharge summary but thin middle will not demonstrate ongoing need. Each day's note should record the current clinical problem, the treatment being delivered that could not be given as an outpatient, and the criteria for discharge that have not yet been met.

Q5. **A coder is told by a manager to change a code so that the case falls into a higher package. What should the coder do?**

**Answer:** Decline to change a code the record does not support, explain that the correct route is a compliant query to the clinician, and if the clinician documents additional clinical detail, code to the updated record. If pressure continues, escalate through the medical records committee or the compliance route, and keep a record of the instruction and the escalation. Changing a code on instruction, without supporting documentation, exposes the coder personally as well as the hospital.

Q6. **Two hospitals treat identical patients. Hospital A records 'type 2 diabetes mellitus with diabetic nephropathy and stage 3 chronic kidney disease, on insulin'; Hospital B records 'diabetic'. Both deliver the same care. What differs?**

**Answer:** Everything downstream. Hospital A's coded data reflects the true complexity, so its expected length of stay and expected mortality are risk-adjusted appropriately, its claim is supported, and its outcome comparisons are fair. Hospital B appears to be treating simple diabetics with unexplained long stays and poor outcomes, and its claims are more likely to be queried. The clinical care is identical; only the record differs, and the record is what everyone else sees.

**ASSESSMENT BANK**

A fuller bank of 50 scored questions for this module is supplied separately as a JSON question bank for loading into your LMS.